

Perspectives on Data Science

Lab sessions



Hösttermin / Autumn 2023

Seven lab sessions, divided into two
subject areas

Tools / Python

Writing / $\text{\LaTeX}$

- Tools for the data scientist (Python)
- Why Python?
- How to install Python

- All in a days work for a data scientist:
 - Get data
 - Manipulate and process data
 - Run simulations, experiments
 - Visualise results

- Lots of high level scientific libraries
 - pandas, numpy, ...
 - tensorflow, Keras, ...
 - matplotlib, seaborn, ...

Introduction

How to install Python

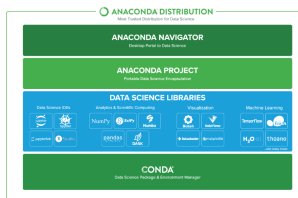
- We use the anaconda distribution
 - Similar on Linux, osx and Windows
 - Includes a number of often used libraries (pandas, &c)
 - Graphical navigator
 - Jupyter notebooks

Download and Install

Anaconda Website



Easily install 1400+ data science packages for Python/R and manage your packages, dependencies, and environments—all with the single click of a button. Free and open source.

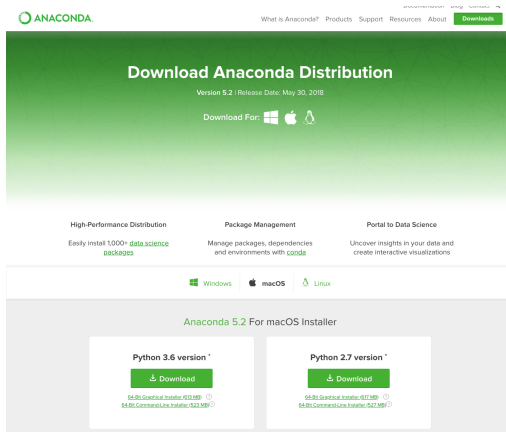


Why Over 6 Million Users Love Anaconda Distribution

<https://www.anaconda.com/distribution/>

Download and Install

Anaconda Downloads



The screenshot shows the Anaconda website's download page. At the top, the Anaconda logo is on the left, and navigation links for 'What is Anaconda?', 'Products', 'Support', 'Resources', 'About', and 'Downloads' are on the right. The main heading is 'Download Anaconda Distribution' with the version '5.2' and release date 'May 30, 2018'. Below this, it says 'Download For:' followed by icons for Windows, macOS, and Linux. Three columns describe the distribution: 'High-Performance Distribution' (easy install of 1,000+ data science packages), 'Package Management' (manage packages, dependencies, and environments with conda), and 'Portal to Data Science' (uncover insights in your data and create interactive visualizations). A section for 'Anaconda 5.2 For macOS Installer' features two download buttons for 'Python 3.6 version' and 'Python 2.7 version'. Each button is accompanied by links to the 64-bit Graphical Installer (637 MB) and the 64-bit Command-Line Installer (527 MB).

ANACONDA

What is Anaconda? Products Support Resources About Downloads

Download Anaconda Distribution

Version 5.2 | Release Date: May 30, 2018

Download For:   

High-Performance Distribution
Easily install 1,000+ [data science packages](#)

Package Management
Manage packages, dependencies and environments with [conda](#)

Portal to Data Science
Uncover insights in your data and create interactive visualizations

Windows macOS Linux

Anaconda 5.2 For macOS Installer

Python 3.6 version *

[Download](#)

[64-Bit Graphical Installer \(637 MB\)](#) ⓘ
[64-Bit Command-Line Installer \(527 MB\)](#) ⓘ

Python 2.7 version *

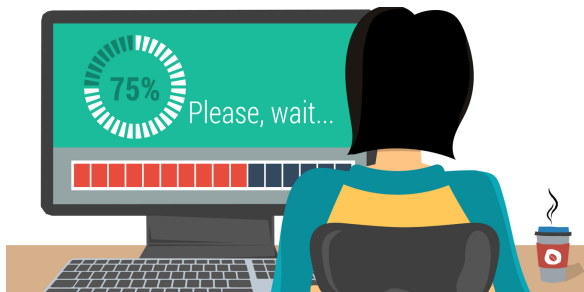
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Download and Install

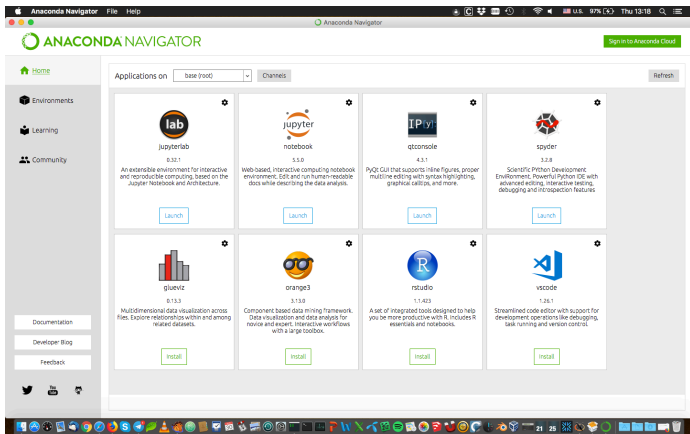
Installing



Open anaconda-navigator

Running

Anaconda-Navigator



Navigator

Perspectives on
Data Science

Before You Click

Getting the Titanic notebook and data set

- Download `Titanic.ipynb` and `titanic.csv` from blackboard
- Save it on your hard disk
- Launch Jupyter Notebook

- A notebook containing live! code, notes and figures
- Information is stored in “cells”
- Run python in an interactive environment (browser)
- server – client model
- Session stored in a notebook file, can be reloaded
- Lots of tutorials on the net use this format
- No messing about on the command line :(
- Great for trying out ideas, data processing, &c

Jupyter Notebook

Overview

Quiz!

In the code box below, write a simple program that calculates how many minutes there are in seven weeks.

```
In [ ]: # insert your code here
```

Cell

Jupyter Notebook

Overview

Quiz!

In the code box below, write a simple program that calculates how many minutes there are in seven weeks.

```
In [ ]: print( 7 * 24 * 60 )|
```

Cell

Jupyter Notebook

Overview

Quiz!

In the code box below, write a simple program that calculates how many minutes there are in seven weeks.

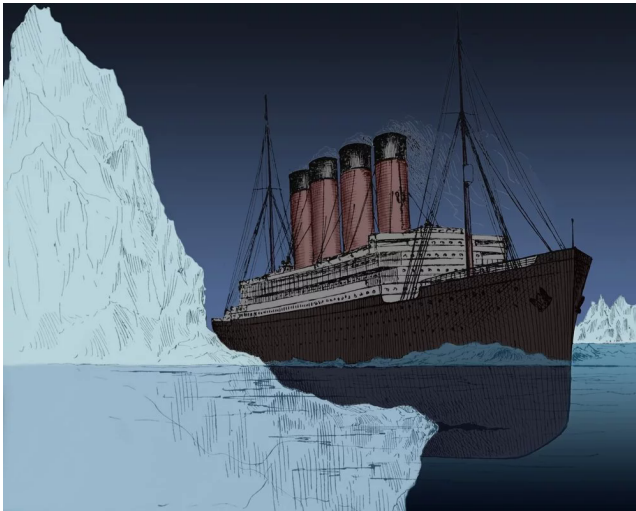
In [1]: `print( 7 * 24 * 60 )`

10080

Cell

Excercises

Titanic



Exercices

Be a data scientist

- Get a data file
- Load it into a pandas data frame
- Get some statistics

Exercises

Titanic

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Select items to perform actions on them.

[Upload](#)[New ▾](#)

☐ 0 ▾	/ Documents / ipynb	Name ▾	Last Modified	File size
☐	..		seconds ago	
☐	Titanic.ipynb		seconds ago	27.3 kB
☐	titanic.csv		seconds ago	98.1 kB

Jupyter Notebook Browser

Titanic

Age	Cabin	Embarked	Fare	Name	Parch	PassengerId	Pclass	Sex	SibSp	Survived
22.0		S	7.25	Braund, Mr. Owen Harris	0	1	3	male	1	0.0
38.0	C85	C	71.2833	Cummings, Mrs. John Bradley	0	2	1	female	1	1.0

Perspectives on Data Science

- Follow the text, fill in the answers in the cells
- Run your code in each cell
- Save the output, and upload it to Blackboard